THE EXTERIOR PRODUCT OF CYCLE CLASSES PASSES TO RATIONAL EQUIVALENCE IN BOTH VARIABLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FBV (sources/stacks-chow/pinned-original-source.tex line 14137).

Allowed standard prerequisites (editorial).

- Flat pullback as a bivariant class, and proper pushforward of bivariant classes. Same chapter, Lemmas labelled lemma-flat-pullback-bivariant (source line 6131) and lemma-push-proper-bivariant (source line 6167), invoked at proof lines 14163-14167. Supply the two operations whose composite c' = i_* o (p')^* caps a class on Y to the exterior product with X'; the identification of the cap products with the cycle products is inside the counted span.
- Cycle groups of schemes over a field and rational equivalence. Same chapter, chow.tex, the definition of Z_n(X) and CH_n(X) for a scheme over a field. Fixes the objects between which the map is defined and the equivalence relation it must respect.

Source reference. Original lemma 0B78: sources/stacks-chow/pinned-original-source.tex, source lines 6131–6131. Same-chapter pointer into the author environment lemma at source line 6131; invoked at proof line 14165 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EPK: sources/stacks-chow/pinned-original-source.tex, source lines 6167–6167. Same-chapter pointer into the author environment lemma at source line 6167; invoked at proof line 14167 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *The map* $\times : \mathrm{CH}_n(X) \otimes_{\mathbf{Z}} \mathrm{CH}_m(Y) \to \mathrm{CH}_{n+m}(X \times_k Y)$ *is well defined.*

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Proof. A first remark is that if $\alpha = \sum n_i[X_i]$ and $\beta = \sum m_j[Y_j]$ with $X_i \subset X$ and $Y_j \subset Y$ locally finite families of integral closed subschemes of dimensions n and m, then $X_i \times_k Y_j$ is a locally finite collection of closed subschemes of $X \times_k Y$ of dimensions $n + m$ and we can indeed consider

$$\alpha \times \beta = \sum n_i m_j [X_i \times_k Y_j]_{n+m}$$

as a $(n+m)$-cycle on $X \times_k Y$. In this way we obtain an additive map $\times : Z_n(X) \otimes_{\mathbf{Z}} Z_m(Y) \to Z_{n+m}(X \times_k Y)$. The problem is to show that this procedure is compatible with rational equivalence.

Let $i : X' \to X$ be the inclusion morphism of an integral closed subscheme of dimension n. Then flat pullback along the morphism $p' : X' \to \mathrm{Spec}(k)$ is an element $(p')^* \in A^{-n}(X' \to \mathrm{Spec}(k))$ by Lemma 0B78 and hence $c' = i_* \circ (p')^* \in A^{-n}(X \to \mathrm{Spec}(k))$ by Lemma 0EPK. This produces maps

$$c' \cap - : \mathrm{CH}_m(Y) \longrightarrow \mathrm{CH}_{m+n}(X \times_k Y)$$

which the reader easily sends $[Y']$ to $[X' \times_k Y']_{n+m}$ for any integral closed subscheme $Y' \subset Y$ of dimension m. Hence the construction $([X'], [Y']) \mapsto [X' \times_k Y']_{n+m}$ factors through rational equivalence in the second variable, i.e., gives a well defined map $Z_n(X) \otimes_{\mathbf{Z}} \mathrm{CH}_m(Y) \to \mathrm{CH}_{n+m}(X \times_k Y)$. By symmetry the same is true for the other variable and we conclude. $\square$